



Abd Alrahman Ahmed Mousa Mohamed

DESKTOP APPLICATION & DATA AUTOMATION DEVELOPER

Egypt · +20 102 964 6085 abdalrahmanahmedmousa@gmail.com github.com/engboda6 Portfolio

EXPERTISE

- Windows desktop applications
- Data analysis automation
- REST API integration
- Interactive visualization
- PDF & Excel reporting

LANGUAGES

ARABIC

Native

ENGLISH

Professional working proficiency

CORE STACK

- React · Electron
- Node.js · Express
- Python · pandas
- SQL · REST APIs
- Git · GitHub

DELIVERY

- Windows installers
- Clean datasets
- Business dashboards
- Client-ready reports
- Technical documentation

PROFESSIONAL PROFILE

Software developer specializing in Electron desktop applications, data analysis automation, and REST API integration. Built production-ready tools for reporting, visualization, and business workflows using React, Electron, Node.js, and Python. Comfortable taking projects from raw requirements and data to packaged Windows software and client-ready deliverables.

SELECTED PROJECTS

Cosmic Observer — Windows Astronomy Application React · Three.js · Electron · Node.js

Problem: Astronomy data is fragmented across multiple scientific sources and difficult to explore visually.

Solution: Built a packaged desktop application with a 3D sky, celestial tracking, astronomy calculations, an AI guide, and integrations with NASA, JPL, CelesTrak, and Vizier.

Result: Unified live astronomy data and interactive visualization in one Windows-ready research experience.

Vertical Farm Telemetry Analysis Python · pandas · NumPy · matplotlib

Problem: Raw farm telemetry required repeatable cleaning, channel-level analysis, and clear operational reporting.

Solution: Created an end-to-end pipeline for preparation, nutrient-decay analysis, weak-circulation detection, charts, Excel output, and PDF generation.

Result: Processed **600 production records**, evaluated **60 irrigation channels**, and generated a **3-page decision report** in one run.

Home Repair Loyalty Analysis Python · Data Analysis · PDF Reporting

Problem: A home-repair business needed actionable loyalty offers from a large marketing dataset.

Solution: Analyzed customer behavior, segmented loyalty opportunities, and translated findings into tiered service bundles and recommendations.

Result: Turned **2,000 customer records** into a structured **5-page client report** with practical retention actions.

TECHNICAL SKILLS

Programming Languages

JavaScript (JSX) Python HTML CSS SQL GLSL

Frontend & Desktop

React 18 Three.js Vite Electron electron-builder

Backend & Integration

Node.js Express REST APIs OpenAI SDK

Data & Analytics

pandas numpy matplotlib openpyxl KPI Analysis

EDUCATION

Computer Science

Faculty of Computers & Information

Ongoing